

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1004	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:		Duration:	60 minutes

Code of Conduct

I A.Sai Joshitha(192424220) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: A.Sai Joshitha

TITLE: ONLINE MOVIE TICKET BOOKING SYSTEM

1. INTRODUCTION

1.1 Background

The Online Movie Ticket Booking System is a web-based application that allows customers to search for movies, view theatres and show timings, check seat availability, select seats, make online payments and receive electronic tickets.

Traditional movie ticket booking requires customers to visit the theatre or wait in queues. It may also be difficult for customers to know the availability of seats and show timings before visiting the theatre.

The proposed system provides a centralized online platform that makes movie ticket booking faster, easier and more convenient.

1.2 Purpose

The purpose of this system is to automate the movie ticket booking process and provide customers with a simple and secure way to reserve movie tickets online.

The system also helps administrators manage movies, theatres, screens, shows, seats, users and bookings efficiently.

2. PROBLEM DESCRIPTION

In the traditional movie ticket booking process, customers may need to visit the theatre physically to purchase tickets. This can result in long queues, increased waiting time and difficulty in checking seat availability.

Manual booking can also result in errors in maintaining records. Customers may not be able to easily compare different theatres and show timings.

The Online Movie Ticket Booking System solves these problems by providing online access to movie information, theatre information, show timings and seat availability.

3. EXISTING SYSTEM

The existing system may use manual or partially computerized ticket booking.

Limitations

- Customers may need to visit the theatre.
- Long queues may occur.
- Seat availability may not be known in advance.
- Manual records can contain errors.
- Cancellation can be inconvenient.

- Customers cannot easily compare different shows.
- Manual management requires more staff effort.

4. PROPOSED SYSTEM

The proposed system provides an online platform for complete movie ticket booking.

Customers can:

- Register and login.
- Search movies.
- View movie details.
- Select theatres.
- Select show timings.
- View available seats.
- Select seats.
- Make online payments.
- Receive booking confirmation.
- Download e-tickets.
- View booking history.
- Cancel eligible bookings.

Administrators can manage:

- Movies
- Theatres
- Screens
- Seats
- Shows
- Users
- Bookings
- Reports

5. OBJECTIVES

The main objectives of the system are:

1. To provide online movie ticket booking.
2. To reduce waiting time and manual work.
3. To provide real-time seat availability.
4. To provide movie and theatre information.

5. To allow customers to select preferred seats.
6. To provide secure online payment.
7. To generate electronic tickets.
8. To prevent duplicate seat booking.
9. To maintain accurate booking records.
10. To provide efficient administration facilities.

6. SCOPE OF THE SYSTEM

6.1 In-Scope

The system includes:

- User registration and login.
- Movie search.
- Movie details.
- Theatre selection.
- Show selection.
- Seat availability.
- Seat selection.
- Ticket booking.
- Online payment.
- E-ticket generation.
- Booking history.
- Cancellation.
- Movie management.
- Theatre management.
- Show management.
- User management.
- Booking reports.

6.2 Out-of-Scope

The following are outside the main scope:

- Movie production.
- Physical theatre construction.
- Physical cash collection.
- Movie content distribution.

- Management of theatre hardware.

7. STAKEHOLDERS

Stakeholder	Role
Customer	Searches movies and books tickets
Administrator	Manages the system
Theatre Manager	Manages theatres and shows
Theatre Staff	Verifies tickets
Payment Gateway	Processes online payments
System Owner	Monitors the system

8. FUNCTIONAL REQUIREMENTS

ID	Functional Requirement
FR01	The system shall allow users to register.
FR02	The system shall allow users to login.
FR03	The system shall allow users to update their profile.
FR04	The system shall allow users to search movies.
FR05	The system shall display movie details.
FR06	The system shall display theatres.
FR07	The system shall display show timings.
FR08	The system shall display seat availability.
FR09	The system shall allow users to select seats.
FR10	The system shall reserve selected seats during payment.
FR11	The system shall calculate ticket cost.
FR12	The system shall process online payment.
FR13	The system shall confirm successful bookings.
FR14	The system shall generate an electronic ticket.
FR15	The system shall display booking history.

FR16	The system shall allow eligible bookings to be cancelled.
FR17	The administrator shall manage movies.
FR18	The administrator shall manage theatres.
FR19	The administrator shall manage shows and screens.
FR20	The administrator shall manage users and bookings.
FR21	The system shall generate booking reports.

9. NON-FUNCTIONAL REQUIREMENTS

9.1 Performance

The system should provide quick responses to user requests and should support multiple users simultaneously.

9.2 Security

The system should protect user accounts and payment-related information. Authentication and authorization should be implemented.

9.3 Reliability

The system should maintain accurate booking information and prevent duplicate seat reservations.

9.4 Availability

The system should be available throughout the day with minimum downtime.

9.5 Usability

The interface should be simple, clear and easy to use.

9.6 Scalability

The system should support increasing numbers of users, theatres, movies and bookings.

9.7 Maintainability

The application should be modular and easy to update and maintain.

9.8 Portability

The system should work on commonly used web browsers and different devices.

10. ACTORS AND CASES

10.1 Customer

The customer is the primary actor.

Use Cases

- Register
- Login

- Search Movie
- View Movie Details
- Select Theatre
- Select Show
- View Seat Availability
- Select Seats
- Make Payment
- Book Ticket
- View E-Ticket
- View Booking History
- Cancel Booking
- Logout

10.2 Administrator

The administrator manages the system.

Use Cases

- Login
- Manage Movies
- Manage Theatres
- Manage Screens
- Manage Seats
- Manage Shows
- Manage Users
- View Bookings
- Generate Reports

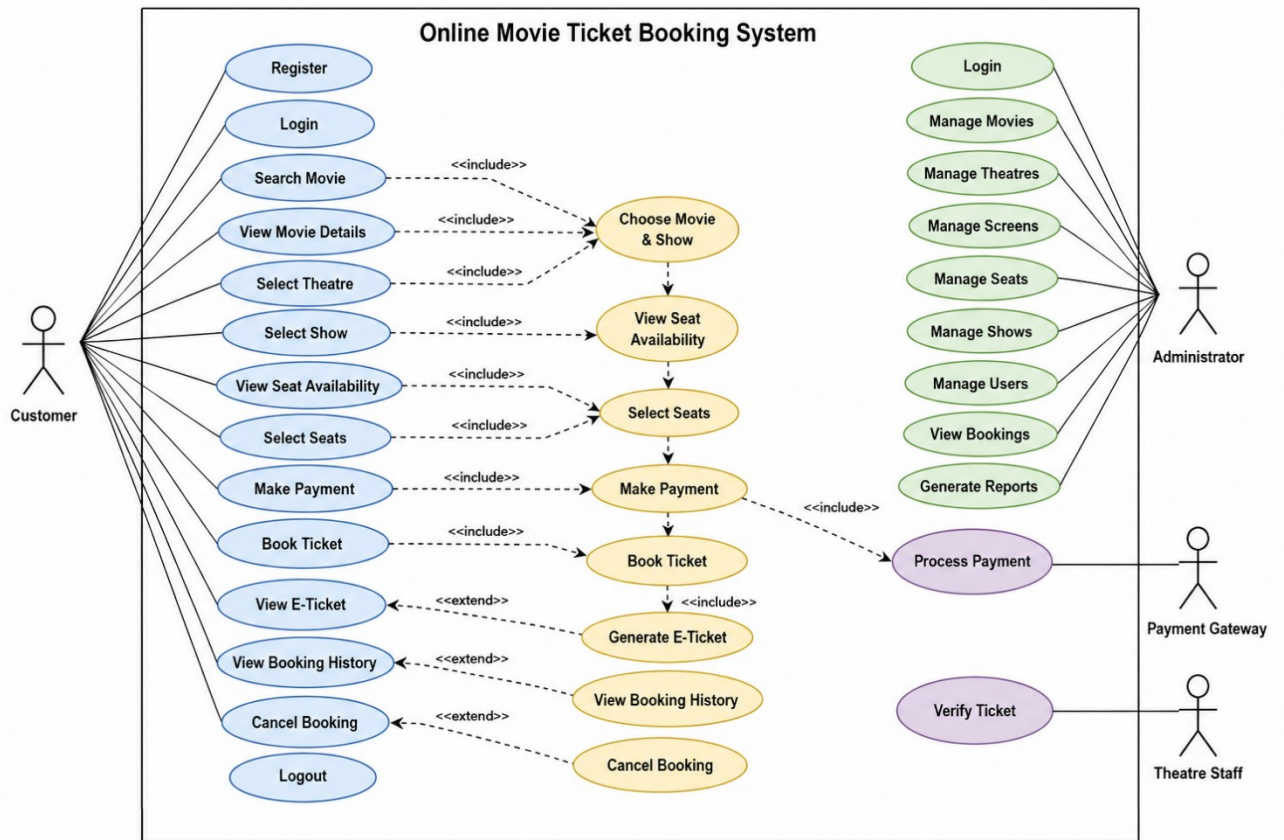
10.3 Payment Gateway

The payment gateway processes the online transaction and returns the payment status.

10.4 Theatre Staff

Theatre staff verify the customer's ticket at the theatre.

11. CASE DIAGRAM



12. CASE DESCRIPTIONS

UC01 – User Registration

Item	Description
Actor	Customer
Precondition	User does not have an account
Main Flow	Enter details → Validate details → Create account
Alternative Flow	Invalid or duplicate details
Postcondition	User account is created

UC02 – Search Movie

Item	Description
Actor	Customer
Precondition	System is available
Main Flow	Enter movie name → Search database → Display results
Alternative Flow	Movie not found

Postcondition	Matching movies are displayed
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UC03 – Select Seats

Item	Description
Actor	Customer
Precondition	Movie and show are selected
Main Flow	Display seat layout → Select available seats → Reserve temporarily
Alternative Flow	Seat unavailable
Postcondition	Selected seats are temporarily reserved

UC04 – Make Payment

Item	Description
Actor	Customer, Payment Gateway
Precondition	Seats are selected
Main Flow	Calculate amount → Select payment method → Process payment
Alternative Flow	Payment failure
Postcondition	Payment status is recorded

UC05 – Generate Ticket

Item	Description
Actor	Customer
Precondition	Payment is successful
Main Flow	Confirm booking → Generate booking ID → Generate e-ticket
Postcondition	Electronic ticket is available

UC06 – Cancel Booking

Item	Description
Actor	Customer
Precondition	Booking is eligible for cancellation
Main Flow	Select booking → Cancel booking → Process refund if applicable
Alternative Flow	Cancellation period expired
Postcondition	Booking status is updated

UC07 – Manage Movies

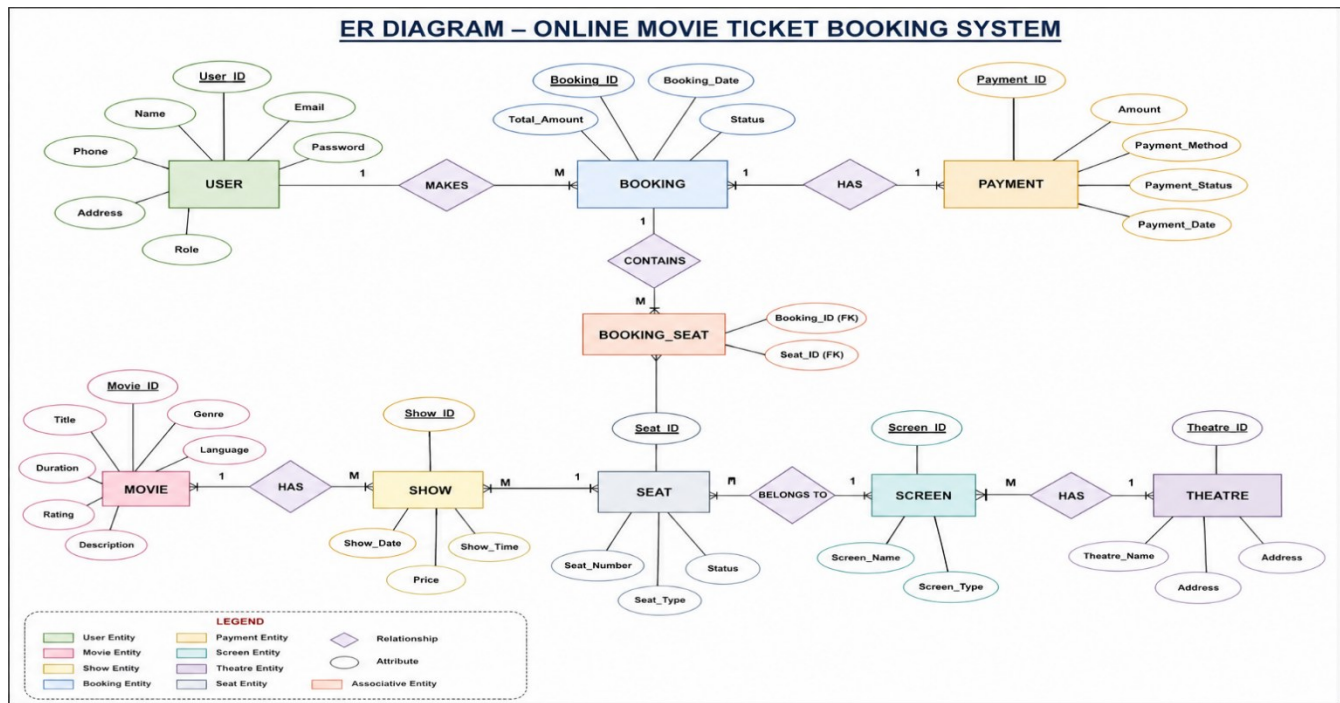
Item	Description
Actor	Administrator
Precondition	Administrator is logged in
Main Flow	Add, update or remove movie information
Postcondition	Movie information is updated

13. DATA REQUIREMENTS

The system stores the following major entities:

Entity	Main Attributes
User	User_ID, Name, Email, Phone, Password
Movie	Movie_ID, Title, Genre, Language, Duration
Theatre	Theatre_ID, Name, Location
Screen	Screen_ID, Theatre_ID, Name
Seat	Seat_ID, Screen_ID, Number, Type, Status
Show	Show_ID, Movie_ID, Screen_ID, Date, Time, Price
Booking	Booking_ID, User_ID, Show_ID, Date, Amount, Status
Payment	Payment_ID, Booking_ID, Amount, Method, Status
Ticket	Ticket_ID, Booking_ID, QR_Code

14. ER DIAGRAM



Relationships

- One User can make many Bookings.
- One Movie can have many Shows.
- One Theatre can have many Screens.
- One Screen can contain many Seats.
- One Show can have many Bookings.
- One Booking has payment information.
- One Booking generates a ticket.

15. EXTERNAL INTERFACE REQUIREMENTS

15.1 User Interface

The system should provide:

- Home page
- Registration page
- Login page
- Movie search page
- Movie details page
- Theatre and show selection page

- Seat selection page
- Payment page
- Booking confirmation page
- E-ticket page
- Booking history page
- Admin dashboard

15.2 Software Interface

The system interacts with:

- Database
- Payment Gateway
- Email/SMS service
- Web browser

15.3 Payment Interface

The system should support online payment methods such as UPI, cards, net banking and digital wallets through a payment gateway.

16. SYSTEM FEATURES

16.1 User Management

Users can register, login, update profiles and logout.

16.2 Movie Search

Users can search movies and view relevant movie information.

16.3 Theatre and Show Selection

Users can select theatres, dates and show timings.

16.4 Seat Selection

The system displays available seats and allows users to select preferred seats.

16.5 Online Booking

The system creates a booking after successful payment.

16.6 E-Ticket

The system generates an electronic ticket containing booking details.

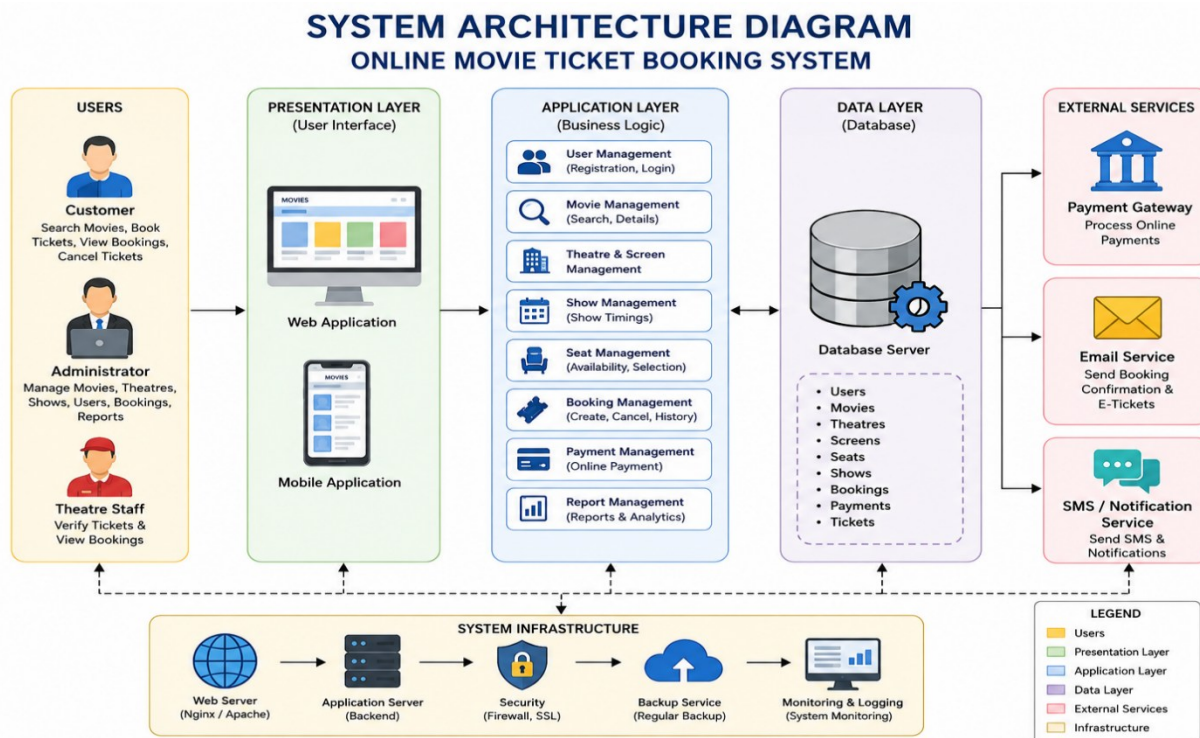
16.7 Cancellation

Users can cancel eligible bookings according to the cancellation policy.

16.8 Administration

Administrators can manage movies, theatres, screens, shows, users and bookings.

17. SYSTEM ARCHITECTURE



Architecture Flow

Customer / Administrator

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Web Application

↓

Application Server

↓

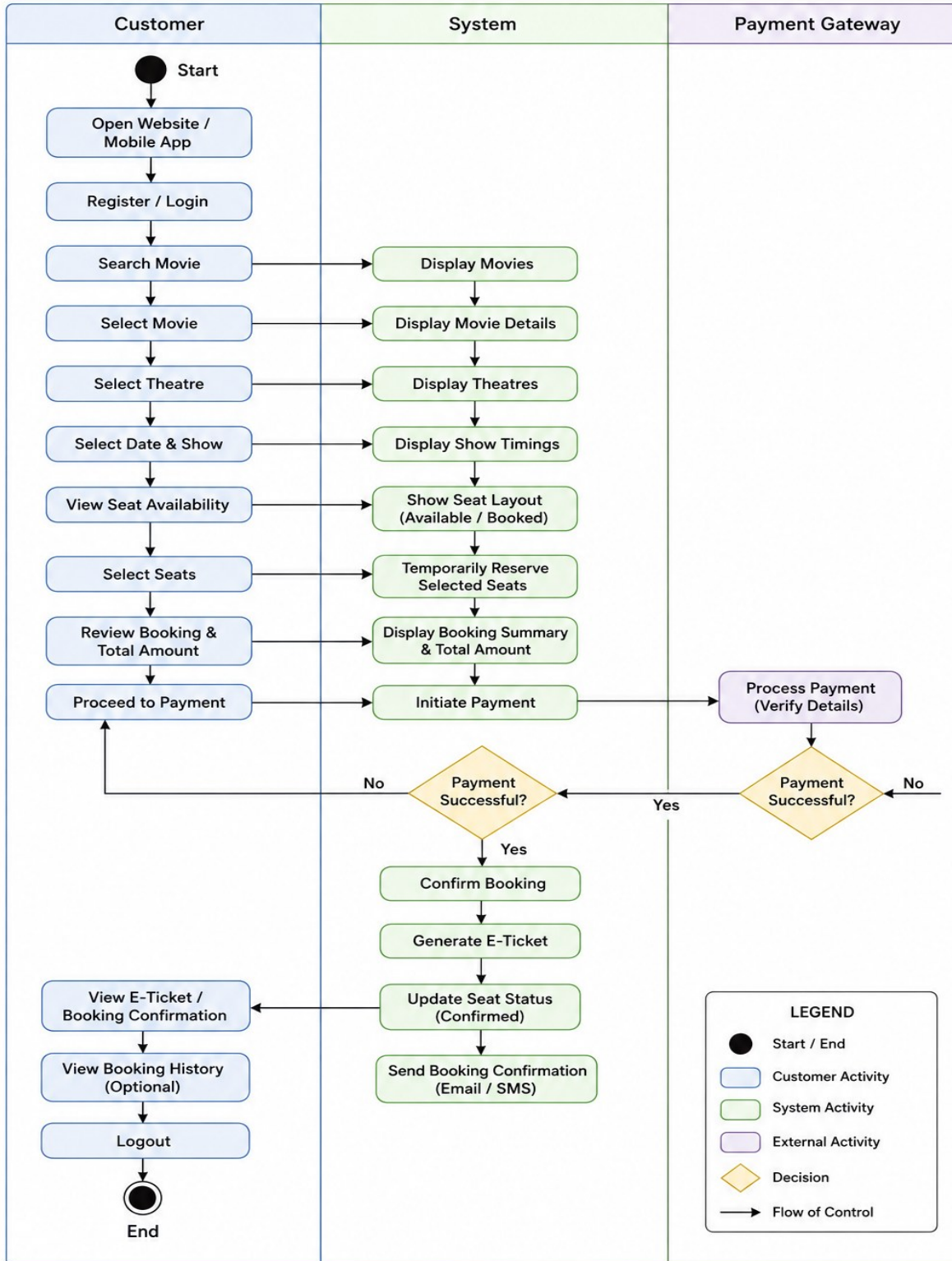
Business Logic

↓

Database

18. ACTIVITY DIAGRAM – BOOKING

ACTIVITY DIAGRAM – ONLINE MOVIE TICKET BOOKING SYSTEM



Activity Flow

Start



Register / Login



Search Movie



Select Movie



Select Theatre



Select Date and Show



View Seat Availability



Select Seats



Confirm Booking



Make Payment

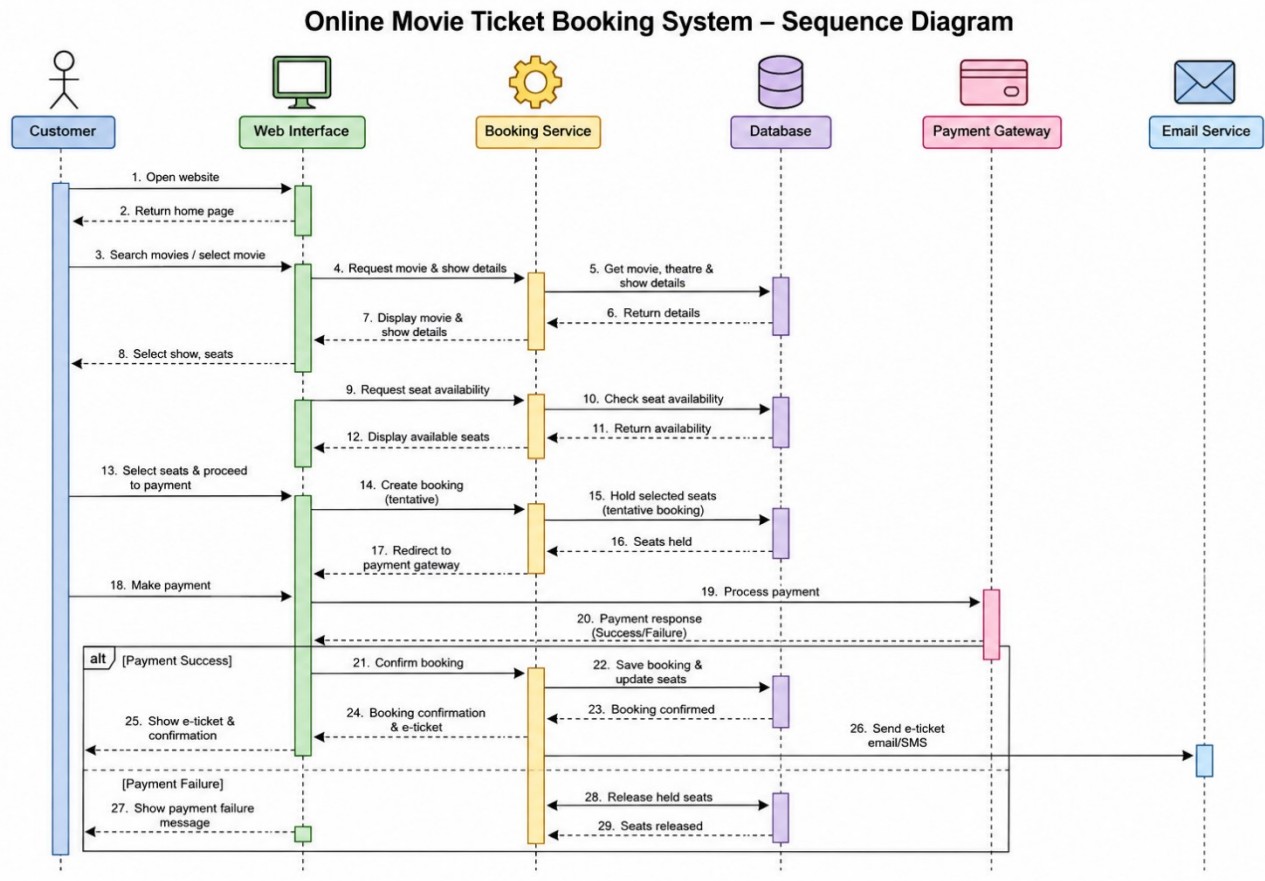


Payment Successful?

If Yes → Confirm Booking → Generate E-Ticket → Display Ticket → End

If No → Display Payment Failure → Retry Payment / Cancel → End

19. SEQUENCE DIAGRAM – BOOKING



Sequence

1. Customer logs into the system.
2. Application validates login information.
3. Customer searches for a movie.
4. Application retrieves movie information from the database.
5. Customer selects theatre and show.
6. Application retrieves seat availability.
7. Customer selects seats.
8. Application temporarily reserves seats.
9. Customer initiates payment.
10. Payment Gateway processes the payment.
11. Payment Gateway returns payment status.

12. Application confirms the booking.
13. Database stores booking details.
14. Ticket Service generates the e-ticket.
15. Application displays booking confirmation.

20. ASSUMPTIONS

1. Users have internet access.
2. Users provide valid information.
3. The payment gateway is available.
4. Administrators maintain accurate movie and show information.
5. Seat availability is synchronized with the booking database.
6. Users have valid payment methods.
7. The database is available during normal operation.
8. Users keep their login credentials confidential.

21. CONSTRAINTS

Technical Constraints

- Internet connection is required.
- Database and server infrastructure are required.
- Payment depends on an external gateway.

Operational Constraints

- Administrators must update movie and show information.
- Theatre staff must verify tickets.
- Users must provide correct information.

Economic Constraints

- Hosting and maintenance have associated costs.
- Payment gateway services may involve transaction charges.

Legal Constraints

- User information must be protected.
- Payment processing must follow applicable regulations.
- Movie information must be used appropriately.

Time Constraints

The project must be completed within the specified academic schedule.

22. REQUIREMENT VALIDATION

Requirements are validated using the following criteria:

Criterion	Result	Explanation
Completeness	Pass	Major system functions are included
Correctness	Pass	Requirements match system objectives
Consistency	Pass	No major conflicting requirements
Non-Ambiguity	Pass	Requirements are clearly written
Non-Redundancy	Pass	Duplicate requirements are avoided
Feasibility	Pass	Requirements can be implemented
Traceability	Pass	Requirements are mapped to use cases

The requirement specification covers the major needs of customers, administrators, theatre staff and payment services.

23. REQUIREMENT TRACEABILITY MATRIX

Requirement	Use Case	Module	Priority
FR01 Registration	UC01	User Management	High
FR02 Login	UC01	Authentication	High
FR04 Search Movie	UC02	Movie	High
FR06 Select Theatre	UC02	Theatre	High
FR07 Select Show	UC02	Show	High
FR08 View Seats	UC03	Seat	High
FR09 Select Seats	UC03	Booking	High
FR12 Payment	UC04	Payment	High
FR14 E-Ticket	UC05	Ticket	High
FR16 Cancellation	UC06	Cancellation	Medium
FR17 Manage Movies	UC07	Administration	High
FR18 Manage Theatre	UC07	Administration	High
FR19 Manage Shows	UC07	Administration	High
FR21 Reports	UC07	Reports	Medium

24. PRODUCT BACKLOG AND PRIORITY

EPIC 1 – User Management

ID	User Story	Priority
US01	As a customer, I want to register so that I can create an account..	High
US02	As a customer, I want to login so that I can securely access the system.	High
US03	As a customer, I want to update my profile.	Medium

EPIC 2 – Movie Search

ID	User Story	Priority
US04	As a customer, I want to search movies.	High
US05	As a customer, I want to view movie details.	High
US06	As a customer, I want to filter movies.	Medium

EPIC 3 – Theatre and Show

ID	User Story	Priority
US07	As a customer, I want to select a theatre.	High
US08	As a customer, I want to select a show timing.	High
US09	As a customer, I want to view available seats.	High

EPIC 4 – Booking

ID	User Story	Priority
US10	As a customer, I want to select seats.	High
US11	As a customer, I want to confirm my booking.	High
US12	As a customer, I want to view booking history.	Medium

EPIC 5 – Payment and Ticket

ID	User Story	Priority
US13	As a customer, I want to make online payment.	High
US14	As a customer, I want to receive an e-ticket.	High
US15	As a customer, I want to cancel eligible bookings.	Medium

EPIC 6 – Administration

ID	User Story	Priority
US16	As an administrator, I want to manage movies.	High
US17	As an administrator, I want to manage theatres.	High
US18	As an administrator, I want to manage shows.	High
US19	As an administrator, I want to manage users.	Medium
US20	As an administrator, I want to generate reports.	Medium

25. CONCLUSION

The Online Movie Ticket Booking System provides an efficient and convenient solution for booking movie tickets online.

The system enables customers to search movies, select theatres and show timings, check seat availability, select seats, make online payments and receive electronic tickets.

The system also provides administrative features for managing movies, theatres, shows, users and bookings.

The requirements identified in this document provide a clear foundation for designing, developing and testing the proposed system. The system improves customer convenience, reduces manual work and provides a reliable method for managing movie ticket bookings.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or
Documentation (15%)	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25